Customer Engagement & Product Utilization Analytics for Retention Strategy

Research paper · European bank customer dataset (10,000 customers) · Prepared by Payal · 26 October 2026

Abstract

Banks often judge churn risk by who a customer is: age, gender, country. This study asks whether how a customer behaves, meaning how active they are and how many products they hold, explains retention better. Using a dataset of 10,000 customers of a European bank, of whom 20.4% left, we classify customers into engagement profiles, measure churn by product depth and balance, and build five retention KPIs and a Relationship Strength Index (RSI).

The main results are: (1) inactive customers churn at 26.9% against 14.3% for active customers; (2) customers with two products are the most loyal (7.6% churn), while customers with three or four products almost always leave (85.9% combined); (3) a high balance does not protect against churn, since funded accounts churn at 22% to 25% regardless of size; and (4) 1,054 high-balance, inactive, single-product customers form an at-risk premium group with 34.3% churn. A model check shows that behavioural variables are about as predictive as demographics (AUC 0.754 vs 0.755), and that combining both lifts AUC to 0.847. Retention strategy should therefore use both, and should target product depth and activity, not balance alone.

1. Introduction and Problem Statement

Retaining an existing customer is usually cheaper than acquiring a new one. Banks hold data on customer activity and product use, yet retention campaigns are frequently generic and misaligned with real behaviour. A customer may look financially strong, with a high balance or salary, and still leave because of low engagement, limited product adoption or a weak relationship with the bank.

The bank lacks three things: quantitative insight into which behaviours drive retention, clarity on whether holding more products reduces churn, and evidence on whether high balances alone ensure loyalty. This project provides that evidence and turns it into a live dashboard.

2. Objectives

Primary objectives

- Evaluate the relationship between engagement and churn.
- Measure the retention impact of product count and product mix.
- Identify disengaged yet high-value customers.

Secondary objectives

- Support engagement-driven retention strategies.
- Improve product bundling decisions.
- Reduce silent churn among premium customers.

3. Data and Exploratory Data Analysis

3.1 Dataset and validation

The dataset has 10,000 rows, one per customer, and 14 columns: CustomerId, Surname, CreditScore, Geography (France, Spain, Germany), Gender, Age, Tenure (years), Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, Exited (churn indicator, the target) and a constant Year column (2025).

Validation results. There were no missing values, no duplicate rows and no duplicate customer IDs. HasCrCard, IsActiveMember and Exited contain only 0 and 1. The churn label is consistent: 2,037 customers exited, a churn rate of 20.37%. Surname and Year carry no analytical value and were not used in analysis.

Profile of churned vs retained customers

Measure	Retained	Churned
Average age	37.4	44.8
Average balance	72,745	91,109
Average tenure (years)	5.0	4.9
Average estimated salary	99,738	101,466
Average credit score	651.9	645.4

Churned customers are older and hold higher balances on average. Tenure and salary barely differ.

3.2 Exploratory data analysis

Numeric variables

Variable	Mean	Median	Std dev	Min	Max	Skewness
CreditScore	650.5	652	96.7	350	850	-0.07
Age	38.9	37	10.5	18	92	1.01
Tenure (years)	5.0	5	2.9	0	10	0.01
Balance	76,486	97,199	62,397	0	250,898	-0.14
EstimatedSalary	100,090	100,194	57,510	12	199,993	0.00
NumOfProducts	1.5	1	0.6	1	4	0.75

Credit score, tenure and salary are roughly symmetric. Age is right-skewed: the median is 37 and 359 customers (3.6%) are flagged as outliers by the interquartile-range rule, mostly older customers up to age 92. These are genuine customers and were kept. Fifteen credit scores are also flagged as low outliers and were kept. Balance and salary have no outliers by the same rule.

Balance has two groups. 3,617 customers (36.2%) hold a zero balance, and the remaining 6,383 have balances centred near 119,800 (standard deviation about 30,100). This is why the high-balance threshold in Section 4 uses non-zero balances only.

Categorical variables

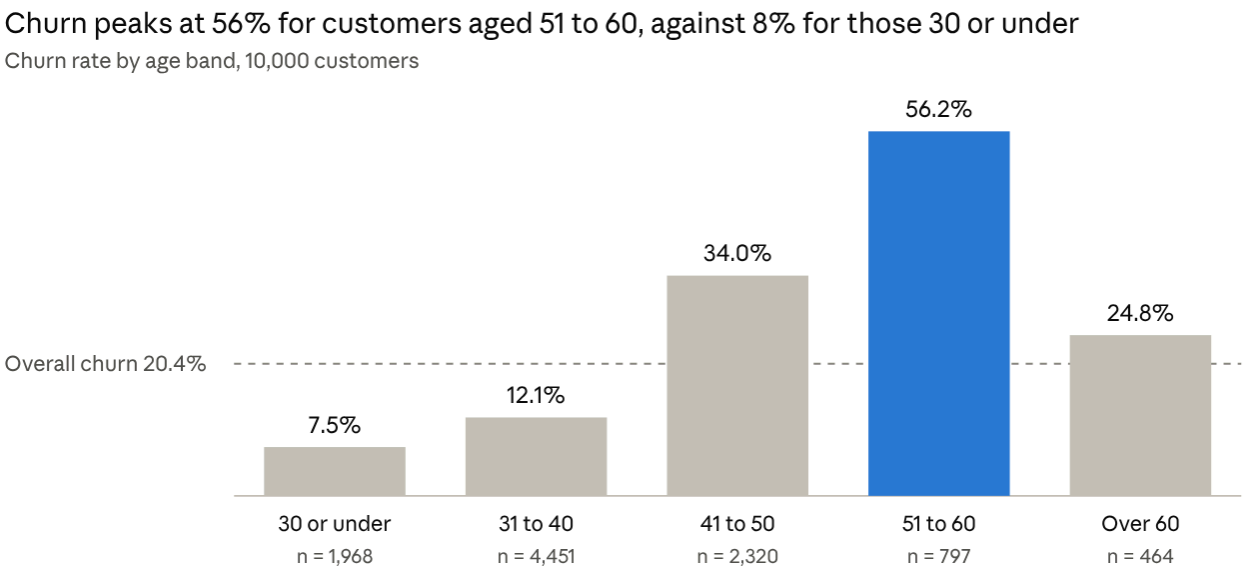
Variable	Distribution
Geography	France 50.1%, Germany 25.1%, Spain 24.8%
Gender	Male 54.6%, Female 45.4%
Products held	1 product 50.8%, 2 products 45.9%, 3 products 2.7%, 4 products 0.6%
Credit card	Has a card 70.6%, no card 29.4%
Activity	Active member 51.5%, inactive 48.5%
Churn (target)	Stayed 79.6%, left 20.4%

The target is moderately imbalanced, about four stayers for each leaver. Three- and four-product customers are rare, which is why their churn rates rest on small groups.

Correlation of each variable with churn

Variable	Correlation with Exited
Age	+0.285
Balance	+0.119
EstimatedSalary	+0.012
HasCrCard	-0.007
Tenure	-0.014
CreditScore	-0.027
NumOfProducts	-0.048
IsActiveMember	-0.156

Age is the strongest linear correlate of churn, followed by activity status and balance. The correlation for product count looks weak, but only because the relationship is not a straight line: churn is low at two products and very high at three or four, so the group-wise analysis in Section 5.2 is needed.



European_Bank.csv · 10,000 customers · age bands as used in Section 5.5

Two data observations that affect interpretation

- **Country and balance are entangled.** Germany has no zero-balance customers, while about 48% of French and Spanish customers do. Funded accounts therefore mix German customers (32.4% churn) with French (18.2%) and Spanish (19.6%) ones. Part of what looks like a balance effect in Section 5.3 is a country effect, and within each country churn does not rise steadily with balance size.

- **Credit score has only a weak link to churn.** Churn is 23.6% for scores of 500 or below and about 20% for scores above 600.

4. Methodology

The analysis followed six steps, implemented in Python (pandas, scikit-learn) and delivered through a Streamlit dashboard.

1. **Data ingestion and validation** as described above.
2. **Engagement classification.** Each customer was placed in one of four profiles, applied in this priority order: *Inactive high-balance* (inactive, balance in the top half of non-zero balances); *Active engaged* (active, two or more products); *Active but low-product* (active, one product); *Inactive disengaged* (all remaining inactive customers). The high-balance threshold uses the median of non-zero balances because 36.2% of customers hold a zero balance, which would distort a plain median.
3. **Product utilization analysis.** Churn rate by number of products, single vs multi-product retention, and a cross-tab of activity by product count.
4. **Financial commitment vs engagement.** Churn by balance band, balance against activity, salary-balance mismatch detection (above-median salary but zero balance) and identification of at-risk premium customers (high balance, inactive, one product).
5. **Retention strength assessment.** A Relationship Strength Index from 0 to 100: 50% activity, 30% product depth (capped at two products, because three or four products do not add strength, see Section 5.2) and 20% tenure. Customers were grouped into four tiers: Weak (0 to 25), Moderate (25 to 50), Strong (50 to 75) and Very strong (75 to 100).
6. **Model check.** Three logistic regression models (demographics only, behaviour only, both) were compared by AUC on a stratified 75/25 train-test split (random state 42) to test whether behaviour adds explanatory power.

5. Findings and Insights

Summary of insights

1. Activity is the clearest behavioural signal: inactive customers leave nearly twice as often as active ones (Section 5.1).
2. Product depth is not linear: two products retain best, three or four products almost always leave (Section 5.2).
3. Balance alone does not signal loyalty, and part of its apparent effect is a country effect (Section 5.3).

4. A group of 1,054 inactive, high-balance, single-product customers has 34.3% churn and holds large balances (Section 5.3).
5. Customers aged 41 to 60 and customers in Germany churn most, and demographics remain important (Section 5.5).
6. Behaviour and demographics are equally predictive alone and much stronger together (Section 5.6).

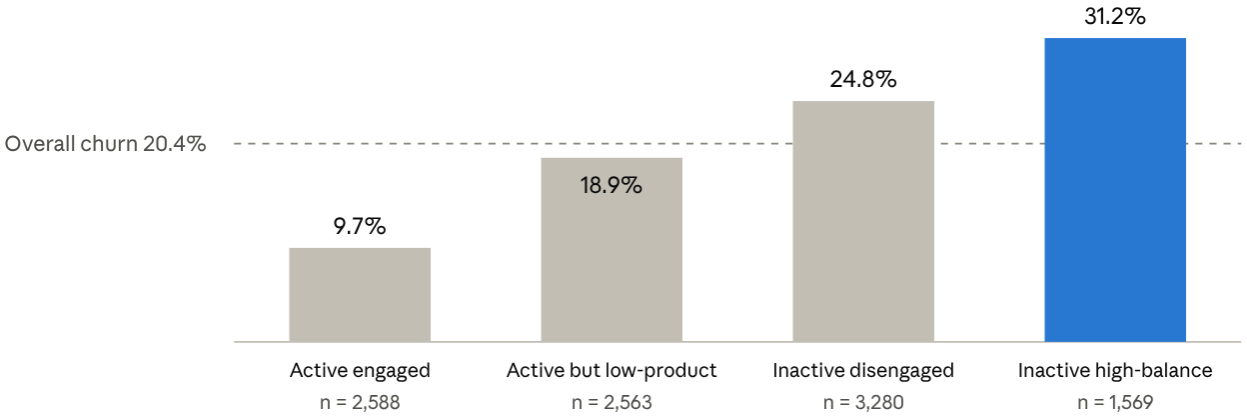
5.1 Engagement and churn

Activity status is strongly related to churn. Inactive members churn at 26.9% (1,302 of 4,849) and active members at 14.3% (735 of 5,151).

Engagement profile	Customers	Churned	Churn rate
Active engaged	2,588	250	9.7%
Active but low-product	2,563	485	18.9%
Inactive disengaged	3,280	812	24.8%
Inactive high-balance	1,569	490	31.2%

Inactive high-balance customers churn at 31.2%, over 3x the active engaged rate

Churn rate by engagement profile, 10,000 customers

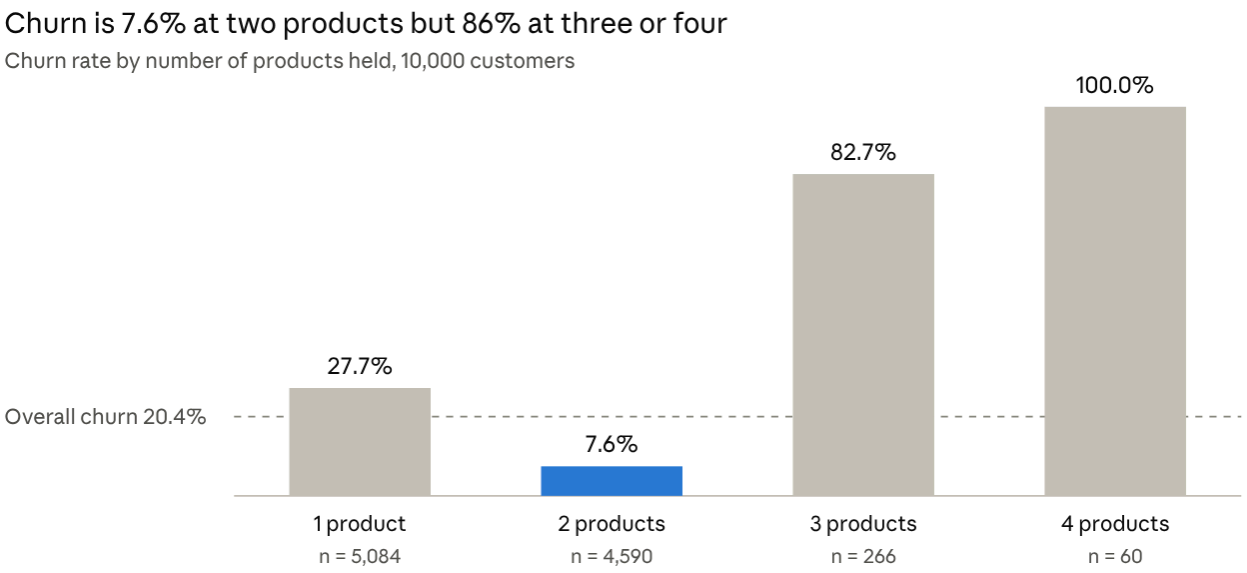


European_Bank.csv · 10,000 customers · profiles as defined in Section 4

The gradient is clear. Customers who are both active and hold multiple products are the most loyal, and customers who are inactive yet hold large balances are the least loyal. The large balance in the last group suggests money is parked rather than committed.

5.2 Product utilization

Products held	Customers	Churn rate
1	5,084	27.7%
2	4,590	7.6%
3	266	82.7%
4	60	100.0%



European_Bank.csv · 10,000 customers · churn = Exited

The relationship is not linear. Moving from one product to two cuts churn by about 20 percentage points, which supports cross-selling a second product. Beyond two products, churn explodes: 280 of the 326 customers with three or four products left. This pattern holds in all three countries (three-product churn of 79% to 90%) and is not rescued by activity: even active members with three or more products churn heavily (inactive 90.2%, active 80.3% for the three-or-more group). Possible explanations are over-selling, product bundles that do not fit the customer, fees, or customers who hold many products because they are already moving their business. The data cannot tell which, so this group needs a qualitative investigation. Because of this result, product depth is capped at two in the RSI.

Single-product customers are loyal only when active. Churn for one-product customers is 36.7% when inactive and 18.9% when active. For two-product customers it is 9.9% and 5.6%.

5.3 Balance, salary and engagement

Balance band	Customers	Churn rate
Zero	3,617	13.8%
Low (under 80k)	585	22.1%
Mid (80k to 120k)	2,617	24.6%
High (over 120k)	3,181	24.1%

A high balance does not ensure loyalty. All funded accounts churn at 22% to 25%, and customers in the top half of funded balances churn at 24.2%, the same as the bottom half (24.0%). Zero-balance customers churn less (13.8%), likely because they are mostly lightly engaged with a single, low-cost relationship.

These pooled figures should be read with care. Germany has no zero-balance customers and a much higher churn rate than France or Spain (Section 3.2), so the country mix inside each balance band affects the comparison.

Balance matters in combination with inactivity. Among inactive customers, high-balance ones churn at 31.2% against 24.8% for lower balances. Among active customers the figures are 17.4% and 12.8%. Disengagement raises the risk, and a large balance raises the cost of losing the customer.

At-risk premium customers. There are 1,054 customers with a high balance, inactive status and a single product. Their churn rate is 34.3%, against 20.4% overall. Of these, 693 had not yet left, holding about 99.2 million in combined balances, which is the pool a retention campaign should target first.

Salary-balance mismatch. 1,777 customers have an above-median salary but a zero balance. Their churn rate is 14.3%, no higher than other zero-balance customers (13.8%), so this mismatch is not a useful warning signal on its own.

5.4 Retention strength

The Relationship Strength Index separates leavers from stayers. Retained customers average an RSI of 60.9 and churned customers 47.5.

RSI tier	Customers	Churn rate
Weak	1,359	37.5%
Moderate	3,490	22.7%

RSI tier	Customers	Churn rate
Strong	1,452	20.0%
Very strong	3,699	12.0%

Churn falls steadily as RSI rises. A sticky customer is active, holds two products and has some tenure. Tenure alone has little effect: churn by tenure year stays within a narrow band of roughly 17% to 23%, so retention strength comes mainly from activity and product depth, not from time with the bank.

5.5 Demographic context

Demographics still matter and should not be ignored.

- **Age:** churn is 7.5% for customers aged 30 or under, 12.1% for 31 to 40, 34.0% for 41 to 50 and 56.2% for 51 to 60, then 24.8% above 60. Customers in their 40s and 50s are the highest-risk age group.
- **Geography:** Germany churns at 32.4% compared with 16.2% in France and 16.7% in Spain. German one-product customers churn at 42.8%.
- **Gender:** female customers churn at 25.1% against 16.5% for male customers.
- **Credit card:** ownership has almost no effect (20.2% with a card, 20.8% without).

5.6 Behaviour vs demographics: model check

Logistic regression model	Variables	AUC
Demographics only	Geography, Gender, Age, CreditScore	0.755
Behaviour only	Activity, products (two vs three-plus), card, tenure, balance, salary	0.754
Combined	All of the above	0.847

Behaviour is as informative as demographics, not more informative. The original hypothesis, that behaviour matters more than demographics, is therefore only partly supported. The strongest conclusion is that the two sets of variables capture different information, and the combined model is much better than either alone. Retention strategy should segment by both.

6. Key Performance Indicators

KPI	Definition	Value
Engagement Retention Ratio	Churn of inactive customers divided by churn of active customers	1.88
Product Depth Index	Churn gap between one-product and two-product customers	20.1 percentage points
High-Balance Disengagement Rate	Share of high-balance customers who are inactive	49.2% (their churn: 31.2%)
Credit Card Stickiness Score	Churn without a card minus churn with a card	0.63 percentage points
Relationship Strength Index	Average RSI of retained vs churned customers	60.9 vs 47.5

The first, second and fifth KPIs show strong signals. The credit card KPI shows almost none, so card ownership is not a meaningful retention lever in this data.

7. Recommendations

1. **Move single-product customers to a second product.** Two products is the point of strongest retention (7.6% churn). Target active one-product customers first, as they are most receptive.
2. **Investigate the three- and four-product group before cross-selling further.** Review fees, product fit and sales practices. Do not push customers beyond two products until the cause of the 86% churn is understood.
3. **Run re-engagement programmes for inactive high-balance customers.** Start with the 693 at-risk premium customers who have not yet left. Offer a relationship review, personalised outreach and a second product suited to them.
4. **Do not rely on balance as a loyalty signal.** Add activity and product count to any retention score used by relationship managers.
5. **Look at Germany and the 40 to 60 age group separately.** Both show very high churn that behavioural measures alone do not explain.
6. **Do not spend retention budget on credit card promotions,** since card ownership has almost no effect.
7. **Use the dashboard as a monitoring tool.** Refresh it with new data regularly and track the five KPIs over time.

8. Limitations

- The data is a single snapshot with no transaction history, so we cannot see when engagement dropped before a customer left. The analysis shows association, not cause.
- IsActiveMember is a binary flag with no definition supplied, and its rule may overlap with churn risk itself.
- The three- and four-product groups are small (326 customers), so their rates are less stable than the rest, although the pattern is consistent across countries.
- Engagement thresholds (the median non-zero balance, the RSI weights) are analyst choices. Different choices would change the group sizes but not the overall pattern.
- The dataset has no currency label or product-type detail, so the product mix cannot be analysed beyond product count.
- The model check uses simple logistic regression. It is meant to compare variable groups, not to be a production churn model.

9. Conclusion

This project reframes customer churn from a behavioural and relationship-strength perspective. Activity and product depth are strong, actionable signals: inactive customers leave nearly twice as often as active ones, and holding two products is associated with the lowest churn. Large balances are not a guarantee of loyalty, and a group of inactive, high-balance customers is quietly at risk. Behaviour and demographics are about equally predictive on their own and much stronger together, so the best retention strategy segments customers by both. The accompanying Streamlit dashboard lets managers filter customers, track the KPIs and export the at-risk list for action.

Appendix: Dashboard modules

The Streamlit application has four modules, each updating with sidebar filters for engagement status, product count, balance, salary and geography: engagement vs churn overview; product utilization impact analysis; high-value disengaged customer detector with a downloadable watch-list; and retention strength scoring panels with the KPI summary.